CODE ARTIFACTS (DESKTOP, WEB AND MOBILE APPLICATION)

PROJECT: Traduce Señas

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Introduction

This document inventories and describes the code artifacts that make up Traduce Señas: where each one is located, how it is organized, which technologies it uses, and how it is built for each target platform (mobile, web and desktop).

The goal is for any evaluator or new team member to be able to locate the source code, understand its structure and generate the executables without relying on verbal explanations.

Project repositories

Table 1
Source code repositories

Repository	URL	Content	Working branch
frontendTS	https://github.com/tadeo77789/frontendTS	React Native / Expo application (mobile, web and desktop)	main
backendTS	https://github.com/mauro-1508/backendTS	API Gateway, five microservices and shared package	main
trans-sl-docs	https://github.com/code-sena/trans-sl-docs	Technical documentation, OpenAPI contracts and prototypes	main

Note. Current branch: main in both repositories, as of November 4, 2026 (backend version 1.4.0, commit a962982; frontend version 1.3.1, commit a5e231d). Changes are merged into main through pull requests using squash and merge.

Client artifact: cross-platform application

Description

The client is a single TypeScript codebase, built with React Native 0.86 and Expo 57, which is compiled for three targets. The react-native-web library allows the same components to run in the browser; files with the .web.ts suffix contain the web-specific implementations (for example, access to MediaPipe in the browser), and Expo selects them automatically depending on the platform.

Table 2
Client targets

Target	Delivery method	User device requirements
Mobile (Android / iOS)	Native application generated with Expo; in development it runs with Expo Go	Android 7.0+ or iOS 15.1+, camera
Web	Static site generated with “expo export --platform web” and served by Nginx in a Docker container	Up-to-date browser with camera access
Desktop	The same web application opened in the browser of a computer (Windows, macOS or Linux), with a layout adapted to screens ≥ 1024 px	Computer with webcam and Chrome, Edge or Firefox browser

Note. No native desktop installer is generated: the desktop version is the responsive web application, which avoids maintaining a fourth codebase.

Folder structure

```
frontend/
├── App.tsx                 entry point
├── app.json                Expo configuration
├── Dockerfile, nginx.conf  build and server for the web version
├── e2e/                    end-to-end tests with Playwright (playwright.config.ts)
└── src/
    ├── app/                config (api.config.ts, i18n), providers, routes
    ├── feature/
    │   ├── auth/           login, registration, verification, passwords
    │   ├── Translation/    translation screen and vision engine
    │   │   └── services/vision/  mediapipeProvider, classifier, normalize,
wordEngine…
    │   ├── Alphabet/       alphabet with 3D models (.glb)
    │   ├── History/        translation history, grouped by sessions
    │   ├── Profile/        profile, terms, privacy, daily challenges and
achievements
    │   ├── Stats/          statistics (administrator)
    │   ├── Admin/          dashboard, training and samples
    │   └── homescreen/     landing page, about us and public demos
    ├── shared/             components, hooks, types, utilities and constants
    └── assets/             fonts, icons, images and alphabet 3D models
```

Main client components

Table 3
Key client files

File	Function
src/app/config/api.config.ts	Base URL of the backend. By default it points to https://traducesenas.tech/api (VPS with Caddy and HTTPS); the EXPO_PUBLIC_API_URL variable changes it at build time, for example to http://localhost:8080/api for a local backend. The production web app uses the relative path /api
feature/Translation/pages/TranslationScreen.tsx	Translation screen with Words / Alphabet selector
feature/Translation/hooks/useSignAgent.ts, useGestureAgent.ts	Capture, recognition and result publication cycle
services/vision/mediapipeProvider(.web).ts	Obtains the 21 hand landmarks in each frame

File	Function
services/vision/normalize.ts	Centers on the wrist and scales by palm size
services/vision/classifier.ts, knnClassifier.ts, motionClassifier.ts	Classification of static letters and signs with movement
services/vision/templateStore.ts, motionTemplateStore.ts	Loading and caching of sign templates
services/translations.service.ts	Stores and queries the history in the backend
feature/auth/services	Real login against the backend and token handling

Note. The current client version is 1.3.1. Each web screen has its own URL; the password routes were renamed. The profile shows the daily challenge progress, the total of challenge points and a queue of notices for completed challenges and unlocked achievements.

Main dependencies

Table 4
Client dependencies

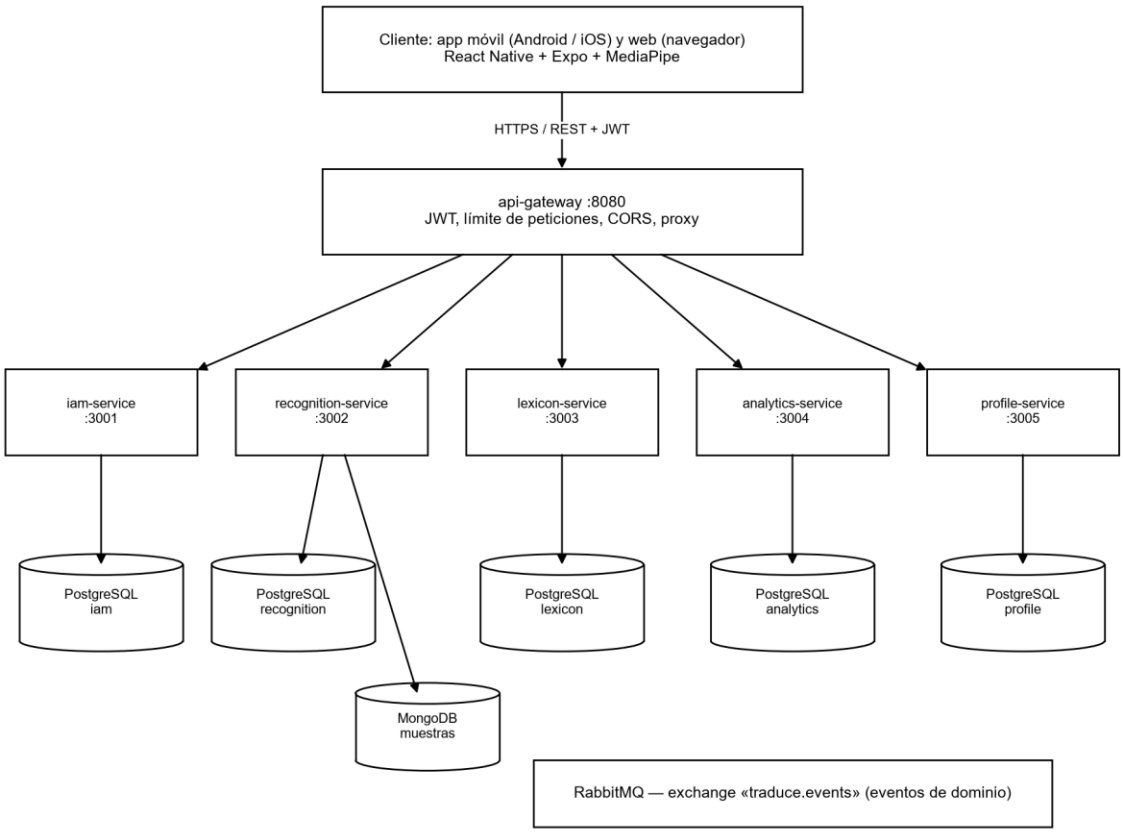
Package	Version	Purpose
expo	57	Cross-platform build platform
react / react-native	19.2.3 / 0.86.3	User interface
react-native-web	0.21	Browser execution
@mediapipe/tasks-vision	0.10.14	Hand detection
@tensorflow/tfjs	4.22	Classification models on the client
expo-camera	57	Camera access
expo-speech	57	Voice output
expo-secure-store / async-storage	57 / 2.2	Secure token and local preferences
@react-navigation/*	6.x	Stack and tab navigation
axios	1.7	HTTP client for the gateway

Server artifact: microservices backend

Microservices architecture

The server is not a single application: it is divided into six artifacts that are built, run and deployed separately. An API Gateway receives all client requests and forwards them to five domain microservices. Each microservice has its own process, its own Dockerfile, its own port and its own database, and none of them reads another's database. The services communicate with each other only through domain events published on RabbitMQ (Newman, 2021).

Figure 1
Backend microservices architecture



Note. Own elaboration.

Table 5
Deployable backend artifacts

Artifact	Responsibility	Port	Own database	Docker image
api-gateway	Single entry point: JWT, rate limiting, CORS and proxy	8080	—	services/api-gateway/Dockerfile
iam-service	Accounts, authentication, roles and permissions	3001	PostgreSQL iam	services/iam-service/Dockerfile
recognition-service	Translations, sign templates and samples	3002	PostgreSQL recognition + MongoDB	services/recognition-service/Dockerfile
lexicon-service	Sign catalog, categories and alphabet	3003	PostgreSQL lexicon	services/lexicon-service/Dockerfile
analytics-service	Usage events and statistics	3004	PostgreSQL analytics	services/analytics-service/Dockerfile
profile-service	Profile, preferences, notifications and achievements	3005	PostgreSQL profile	services/profile-service/Dockerfile

Note. The @traduce/shared package is not a service: it is a common library (configuration, JWT,

middleware and event bus) that each microservice includes in its own image.

Communication between microservices

Table 6
Communication mechanisms

Type	Mechanism	Example
Client → backend	REST over HTTPS through the api-gateway, with JWT	POST /api/translations → recognition-service
Service → service	Asynchronous events in RabbitMQ (traduce.events exchange)	recognition.TranslationProduced → profile and analytics
Shared data	Logical identifiers only (user_id), with no foreign keys between databases	translations.user_id references iam without FK

Structure of the service repositories

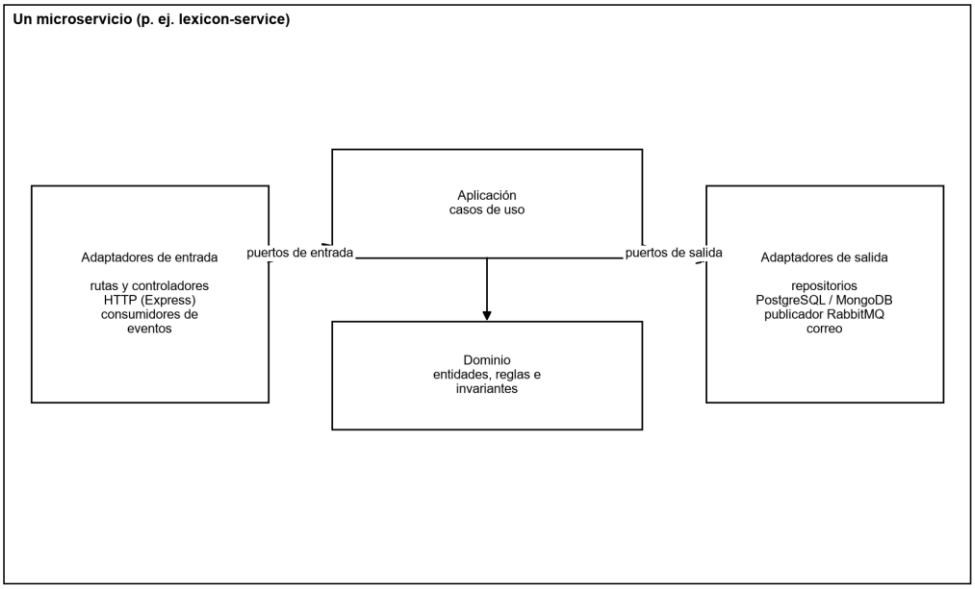
The six artifacts are stored in a single repository to share build tools, but each one lives in its own folder with its package.json, its Dockerfile, its migrations and its tests:

```
backend/
├── docker-compose.yml      orchestrates the 6 services, 5 PostgreSQL, MongoDB and
RabbitMQ
├── packages/shared/        common library @traduce/shared
└── services/
    ├── api-gateway/        Dockerfile · package.json · src/ · tests/
    ├── iam-service/        Dockerfile · package.json · db/ · src/ · tests/
    ├── recognition-service/ Dockerfile · package.json · db/ · src/ · tests/
    ├── lexicon-service/    Dockerfile · package.json · db/ · src/ · tests/
    ├── analytics-service/  Dockerfile · package.json · db/ · src/ · tests/
    └── profile-service/    Dockerfile · package.json · db/ · src/ · tests/
```

Internal structure of each microservice

Within each service, hexagonal architecture is applied: the domain and the use cases do not depend on Express or the database, but on ports implemented by adapters.

Figure 2
Hexagonal architecture within a microservice



Note. Own elaboration.

```
services/lexicon-service/
├── Dockerfile
├── db/003_lexicon.sql          migrations of its own database
└── src/
    ├── domain/                 entities and rules
    ├── application/            use cases
    ├── ports/                  interfaces
    ├── adapters/inbound/http/  Express routes
    ├── adapters/outbound/      PostgreSQL repository and event publisher
    ├── lexicon.module.ts       dependency composition
    └── app.ts                  service startup
```

Inventory and code size

Table 7
Code size by component

Component	.ts files (src)	Lines (src)	Test files
api-gateway	7	265	3
iam-service	77	2 094	33
recognition-service	58	3 016	17
lexicon-service	23	1 629	5
analytics-service	13	487	6
profile-service	20	823	27
@traduce/shared	21	671	7
Total backend	**219**	**8 985**	**99**
Client (frontend/src, .ts and .tsx)	109	16 713	—

Note. The .ts files of src and their lines come from the count of October 7, 2026. The test files come from

the count of November 4, 2026 on the main branch (91 in the services, 7 in @traduce/shared and 1 in the root tests folder).

Exposed interfaces

Table 8
Public API Gateway routes

Prefix	Service	Operation examples
/api/auth	iam-service	register, login, verify-email, resend-verification, forgot-password, reset-password
/api/users, /api/iam	iam-service	me, me/password, roles, role assignment to users
/api/translations	recognition-service	Save translation, history, logical deletion
/api/sign-templates, /api/samples, /api/recognition	recognition-service	Sign templates and training samples
/api/lexicon	lexicon-service	alphabet, search, categories, signs, publish, locations, resources
/api/analytics	analytics-service	events, summary, daily, top-signs, reports/sections
/api/profile, /api/notifications, /api/achievements	profile-service	me, me/stats, notifications, achievements, challenges/today and challenges/rewards/summary (daily challenges and challenge points)

Note. Complete OpenAPI contracts in 07-api/contracts/openapi of the documentation repository.

Building the artifacts

Backend: one artifact per microservice

Each microservice is built, tested and deployed independently:

```
cd backend
cp .env.example .env                                  # JWT_SECRET (32+) and passwords
npm install
npm run build -w @traduce/lexicon-service              # build a single service
npm test                                               # tests for all services
docker compose build lexicon-service                   # single-service image
docker compose up -d --no-deps lexicon-service         # deploy only that service
docker compose up --build -d                           # start all services
```

Mobile application

```
cd frontend
npm install
npx expo start --android     # or --ios; opens in Expo Go
# To generate the installer (APK/AAB or IPA), EAS Build is used:
npx eas build --platform android
```

The EAS Build configuration is in the eas.json file of the client repository.

Web and desktop application

```
cd frontend
npx expo export --platform web --output-dir dist
docker build -t traduce-senas-web .
docker run -p 8081:80 traduce-senas-web
```

Code conventions and version control

- Strict TypeScript on the client and server.
- Backend file names in snake_case; React components in PascalCase.
- Commits in English following Conventional Commits (feat, fix, chore, docs, test, refactor).
- One branch per feature; work is never done directly on main.
- Pull request with review and squash and merge.
- Sensitive variables only in .env (excluded by .gitignore); never in the code.

Conclusions

The project's code artifacts are a single client delivered on mobile, web and desktop, and a microservices backend made up of an API Gateway and five domain services, each with its own Docker image and its own database. Each microservice can be built, tested and deployed without affecting the others, and Docker Compose makes it possible to start all of them together when the complete system is needed.

References

Broadcom. (2026). *RabbitMQ documentation*. https://www.rabbitmq.com/docs

Docker Inc. (2026). *Docker Compose documentation*. https://docs.docker.com/compose/

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Martin, R. C. (2017). *Clean architecture: A craftsman's guide to software structure and design*. Prentice Hall.

Newman, S. (2021). *Building microservices: Designing fine-grained systems* (2nd ed.). O'Reilly Media.

Zhang, F., Bazarevsky, V., Vakunov, A., Tkachenka, A., Sung, G., Chang, C.-L., & Grundmann, M. (2020). *MediaPipe Hands: On-device real-time hand tracking* (arXiv:2006.10214). arXiv. https://doi.org/10.48550/arXiv.2006.10214